

SCADA Trainer

1. Problem statement

What problem does this solve? For whom? How do you know this is real?

SCADA (Supervisory Control And Data Acquisition) computer systems are critical pieces of industrial process supervision and maintenance schemes, and come with all the technical complexity and exacting specifications expected thereof. Training on these critical systems is difficult to obtain if you do not already have prior experience or if you do not work for a large industrial operations company that has already created a training tool.

This SCADA Simulator reduces the barrier of training, allows for practical training without repercussion, and focuses specifically on students in power grid maintenance programs who do not have a suitable alternative to practice their learning, as none currently exists.

2. Target user

Be specific. Demographics, context, and what they currently do instead of using this product.

The target user is a student at a small to mid-size institution of education who is learning power systems management and operation or a related field. They are learning how to operate SCADA computer systems and have no practical means to test their learning, which is theoretical or reliant upon expensive tools that are not built for this purpose.

Another possible user is someone who has been hired to a company that uses SCADA systems but does not have the budget to develop its own bespoke SCADA tooling and does not want to bear the expense of market-standard SCADA training tools.

3. MVP scope

List the 3–5 features that define the minimum viable product. What is explicitly out of scope for V1?

The main requirements for the MVP comprise technical and visual aspects.

On the technical side the app must simulate the IEEE 14-Bus system, a common test case representing a traditional American power system; it must allow the user to interact with the grid using a panel of controls; it must include alarms for alerting the user of simulated warnings or alerts; and it must include at least one test case to allow the user to practically experience the operation of a SCADA system using the alarms and control panel.

Visually the app must render all of these components in a clear and consistent way using color-coding to denote severity of alerts.

Frequency visualization is not a part of the MVP, nor is packaging the simulator as a downloadable app: to begin, it will be hosted in the cloud and accessible on the web.

4. Business model

Revenue streams, pricing model, or sustainability strategy. Who pays, how much, and why?

The software will be offered on a subscription basis (SaaS) with a free trial available before purchase. Pricing is to be determined, with a period of subscription of one year.

Under consideration is both the sale of software licenses directly to students as well as service contracts directly with academic departments. The former sales option reduces costs for students who do not want to use the tool while providing strong value for those that do, and the latter sales option allows for the sale of a flexible amount of licenses directly to a single institutional buyer on behalf of many users.

5. Success metrics

2–3 measurable outcomes that indicate the MVP is delivering value.

Feedback directly from users (individual users or institutional/departmental purchasers) will send a strong signal for the MVP's alignment to the market.

Usage statistics from the cloud hosting provider can act as implicit feedback, providing data on usage metrics in both an individual and aggregated basis.

6. Key risks

Top 2–3 risks (business, technical, or market) and one mitigation idea each.

The largest risk is of technical complexity: there are no electrical engineers on the development team, which has had to learn how to model the complex electrical math for this project. Bringing an expert onto the team would assist with validating and increasing the accuracy of the current code.

The second risk involves market data, or lack thereof: the development of this project relies upon the assumption that there is a market for this tool, which has been anecdotally confirmed but includes no data to confirm veracity of this assumption. Research of the expected market and its size at educational institutions and power companies can confirm or redefine market assumptions.